

SAARTHI

■■■■■ - *the charioteer who guides*

Predictive personnel stress & welfare monitoring for uniformed forces

PS ID

26186

Org

CRPF / Ministry of Home Affairs

Theme

Software | MedTech / HealthTech

Rule

Welfare, not discipline

The problem

Stress is caught after incidents - not before.

654

CAPF suicides in 5 years

~50,000

resignations in the same window

Source: ThePrint, 654 suicides, 50,000 resignations in 5 years - the crisis stalking Indias CAPFs. One statistic. Cited.

Today

Manual observation + self-report. Help arrives late.

Stigma

A named mental-health app is dead on arrival in ACR culture.

Need

Predictive welfare, privacy in architecture, not in a promise.

Team & roles

Six members. Each answers their own domain. ≥ 1 female member (SIH rule).

kv

PM | backend

Engines, privacy law, architecture

neel

Co-backbone

Apps, dashboards, design, RBAC

ayush

ML assistant

Citations, instrument factsheet

manan

Compliance

Statute pack, privacy walkthrough

risa

Presentation

Official-template deck + practice

tejas

QA / demo

Manual tests, fallback video

Solution in one loop

1

Detect

HR signals the force already holds.
Zero extra work.

2

Explain

Rules engine. Top-3 factors. No
black box.

3

Help

Buddy -> counsellor <= 24 h ->
roster change -> Tele-MANAS.

4

Verify

Trend down. Who-viewed receipt.
Loop closed.

THE FIREWALL

An individual risk score cannot reach command, ACR, promotion or posting. Not a policy sentence - there is no API that can carry it there.

How help is routed

Grandma test: jawan keeps doing duty -> counsellor gets an explainable alert in 24 hours -> commander never sees a name.

Green	0 - 39	Self-help nudge	App
Amber	40 - 59	Buddy + informal JCO check	72 h
Red	60 - 79	Counsellor outreach	<= 24 h
Critical	80+	Immediate contact + duty mod	Now

What we built vs what we used

Built	Off the shelf	Deliberately not
Rules engine v1 (explainable)	FastAPI + PostgreSQL / SQLite	No ML accuracy claims (no labels)
HR signal features (20)	Expo / React Native (apps)	No iOS in v1
k-anonymity ≥ 5 aggregator	Next.js consoles	No blockchain / IoT combo
Dual-key unmask + who-viewed	On-device voice features only	No foreign SaaS on welfare data
Append-only hash-chained audit	Tele-MANAS 14416 recorded handoff	On-prem / MeghRaj deployable

Two-tier output - the split is the product

COUNSELLOR

Sees a person

Pseudonym until dual-key unmask

Score + top-3 factors

Consent artefact required

Every read lands in who-viewed

COMMANDER

Sees a unit

Aggregates only, $k \geq 5$

No names, no scores, no IDs

Personnel lookup returns 403

Complement cells also suppressed

Why not a consumer wellness app?

Consumer / HRIS		SAARTHI
Privacy	Policy promise	Architectural firewall (k >= 5)
Offline	Cloud-first	Low-end Android, queue + sync
Language	English UI	Hindi + English, icon-first
Care path	App content	Tele-MANAS 14416 recorded
Sovereignty	Foreign SaaS	On-prem / NIC MeghRaj
AI claim	Black-box %	Rules v1; ML only with labels

What works today

Backend on main. Demo persona: Constable, 34, 3rd Bn. Password for all seed users: saarthi.

GET /risk/ps_demo01/explanation

Counsellor sees factors, not a black box

GET /aggregates/unit/3BN

Commander sees k >= 5, no names

GET /welfare/personnel/{id}

403 - the judge's attack, pre-answered

GET /app/who-viewed

Jawan sees who opened the record

36-hour finale - hours attached

0-1

kv + neel

Clone, seed, pytest, API up

1-4

neel / kv

Wire apps; freeze ruleset

4-8

risa + tejas

Venue deck + new fallback video

8-12

kv + neel

One bug pass (offline, k, dual-key)

12-16

rotation

Sleep. 3 up / 3 down.

16-24

all

Mentor comments -> board. 3x rehearsal.

24-32

neel / kv

UI polish only. Q&A drill.

32-36

tejas

Bag: laptop, hotspot, pen drive, PDF, mp4

Pilot KPIs - no invented numbers

$\geq 30\%$

Voluntary participation

Reasoned estimate (roster-first)

≤ 24 h

Red flag -> counsellor

Design target (FR-09)

0

Scores reaching command

Architectural guarantee

0

Punitive outcomes

Monitored trust signal

SCALE (REASONED, NOT ADJECTIVES)

Technical High - rules recompute a battalion on one box. Adoption Med - needs officer-first buy-in. Outcome evidence Low->Med - labels accrue over months.

What we will not build

Reads as maturity. Preempts: what will you NOT build?

No facial emotion / CCTV

Contested science. Reads as surveillance.

No phone or relationship monitoring

Trust-killing. Fails DPDP necessity.

No scores in ACR / promotion / posting

Firewall is architectural.

No ML on day one

Zero labels. Honest rules instead.

No clinical diagnosis

Reflection support, not a condition.

No automated weapon restriction

Counsellor recommends. Commander decides.

THE NUMBER TO REMEMBER

0

individual risk scores that can reach command. Guaranteed by architecture - not by a promise.

Repo `github.com/dyrok/sih-hackathon-26` | Demo `python -m app.seed && uvicorn app.main:app`

Welfare, not discipline.